

Qwen3.5-4B EN: gender — every mitigation against the baseline on the same rows

| Mitigation               | Condition | Unstable sets %<br>[95% CI] | Δ pp  | Fixed :<br>broken | p     | MAD<br>pp ↓ | AR range<br>pp ↓ | Cohen's h<br>↓ | Inconsist.<br>% ↓ | Hire<br>%     | Utility<br>% ↑ | Mean<br>FS  | Attr.<br>mentioned % |
|--------------------------|-----------|-----------------------------|-------|-------------------|-------|-------------|------------------|----------------|-------------------|---------------|----------------|-------------|----------------------|
| baseline (no mitigation) | implicit  | 9.11 [6.22, 12.00]          |       |                   |       | 0.79        | 3.78             | 0.02           | 2.10              | 22.77         | 80.72          | 0.69        | 0.43                 |
| SFT (LoRA, merged)       | implicit  | 5.33 [3.33, 7.56]           | -3.78 | 34 : 17           | 0.02  | 0.49 (0.79) | 2.22 (3.78)      | 0.01 (0.02)    | 0.87 (2.10)       | 21.71 (22.77) | 81.56 (80.72)  | 0.66 (0.69) | 0.34 (0.43)          |
| baseline (no mitigation) | explicit  | 12.44 [9.11, 15.78]         |       |                   |       | 1.10        | 6.44             | 0.03           | 2.52              | 33.46         | 81.97          | 0.69        | 0.53                 |
| SFT (LoRA, merged)       | explicit  | 6.00 [3.56, 8.67]           | -6.44 | 49 : 20           | 6e-04 | 0.50 (1.10) | 2.22 (6.44)      | 0.02 (0.03)    | 1.09 (2.52)       | 24.33 (33.46) | 82.44 (81.97)  | 0.66 (0.69) | 0.32 (0.53)          |

Values: mitigated run, baseline in brackets, both on exactly the rows the run evaluated. Unstable sets: share with its own 95% CI (bootstrap over candidates); Δ pp: change against the baseline; bold coloured Δ = p < 0.05 (exact sign test on fixed vs broken sets, uncorrected). A run with parse failures is compared on its complete sets only; its own baseline on those sets is shown under its value. Italic ♦ = best value in the column within its condition (real mitigations only; hire rate and FS have no better direction).

Fixed : broken — each counterfactual set (one candidate–job pair, one condition, all attribute variants) paired with the same set at baseline. Fixed: unstable at baseline (the attribute alone changed the decision), stable after the mitigation. Broken: stable at baseline, unstable after — the mitigation introduced a new dependence on the attribute. Δ unstable sets = (broken – fixed) / sets; e.g. 58 : 57 means the method fixes about as many sets as it breaks.

\* Lexical scrubbing is an oracle upper bound, not a comparable mitigation: it removes the attribute with the study's own injection templates, so every variant becomes character-for-character the attribute-free CV (100% of rows) and the model sees one identical prompt — zero instability by construction. Real CVs do not use our templates; read this row only as the ceiling. LLM scrubbing, which must find the attribute itself, is the realistic version.